

David Dooley

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EDUCATION

University of Tennessee

Knoxville, TN

2019–2026

Degree: Ph.D. Chemical Engineering

GPA: 3.94

University of Tennessee

Knoxville, TN

2019–2022

Degree: M.S. Chemical Engineering

GPA: 3.94

Tennessee Technological University

Cookeville, TN

2015–2019

Degree: B.S. Chemical Engineering

GPA: 3.87

EXPERIENCE

Postdoctoral Researcher

University of Tennessee

May 2026–Present

Graduate Research Assistant

University of Tennessee

Aug. 2019–May 2026

- Developing safe and effective CRISPR-Cas antimicrobials against virulent pathogens.
- Engineering stable and specific microbial production platforms for CRISPR-carrying phage-like particles.
- Understanding transcriptomic and proteomic response of target pathogens to treatment with CRISPR-Cas antimicrobials.

Teaching Assistant

University of Tennessee

Aug. 2020–Dec. 2023

- Served as TA for the following courses:
 - Fall 2020: CBE 350 - Chemical and Bioengineering Reactor Fundamentals
 - Spring 2021: CBE 340 - Mass Transfer and Separation Processes
 - Spring 2022: CBE 551 - Advanced Chemical and Biomolecular Engineering Kinetics and Reactor Design
 - Fall 2023: CBE 350 - Chemical and Bioengineering Reactor Fundamentals
- Graded homework and exams and proctored tests.
- Held office hours to provide instruction and feedback on assignments and grading.

REU Research Assistant

National Science Foundation

May 2018–Aug. 2018

- Investigated effect of feedstock pretreatment on kinetics of cellulosic pyrolysis.
- Designed, performed, and presented various experiments involving thermogravimetric, rheological, and X-ray diffraction-based analyses.

PUBLICATIONS

- **Dooley, D.**, Carper, D. L., Giannone, R. J., & Trinh, C. T. (2025). Proteome Reallocation Reveals Dynamics and Mechanisms of Phage Infection and Defense in *Staphylococcus aureus*. bioRxiv. (*in review*)
- **Dooley, D.**, Ryu, S., Giannone, R.J., Edwards, J., Dien, B.S., Slininger, P.J., & Trinh, C.T. (2024). Expanded genome and proteome reallocation in a novel, robust *Bacillus coagulans* strain capable of utilizing pentose and hexose sugars. mSystems, e00952-24.
- Mendoza, B.**, Fry, T.**, **Dooley, D.****, Herman, J., & Trinh, C. T. (2022). CASPER: An Integrated Software Platform for Rapid Development of CRISPR Tools. The CRISPR Journal, 5(4), 609-617.
- Walker, C., Ryu, S., Garcia, S., **Dooley, D.**, Mendoza, B., & Trinh, C. T. (2022). Gene Coexpression Connectivity Predicts Gene Targets Underlying High Ionic-Liquid Tolerance in *Yarrowia lipolytica*. mSystems, 7(4), e00348-22.
- **Dooley, D.**, Trinh, C.T. Simultaneous Cotargeting of Essential and Defensive Gene Targets in *Staphylococcus aureus* Increases Potency and Mitigates Resistance to CRISPR Antimicrobials. (*in preparation*)
- **Dooley, D.**, Trinh, C.T. Improving the Specificity and Host Range of a Phage-Based Delivery System in *Staphylococcus aureus*. (*in preparation*)
- **Dooley, D.****, Boyd, H.**, & Trinh, C. T. Systematic Analysis of Multitargeting Guide RNAs in *Staphylococcus aureus* for Next-Generation CRISPR Antimicrobials. (*in preparation*)

** equal contribution

PRESENTATIONS

- **Dooley, D.**, Trinh, C. T. "Assessing Essential and Defensive Gene Co-Targeting to Enhance CRISPR Antimicrobial Efficacy Against *Staphylococcus aureus*." 2025 AIChE Annual Meeting. Boston, MA. 11/6/25.
- **Dooley, D.**, Ryu, S., Edwards, J., Ha, K., Giannone, R., Dien, B., Trinh, C. T. "Elucidating the Robustness of *Bacillus coagulans* B-768 for Biomanufacturing: Insights from Phenotypic Profiling, Functional Genomics, and Proteome Reallocation." 2025 AIChE Annual Meeting. Boston, MA. 11/6/25.
- Boyd, H., **Dooley, D.**, Trinh, C. T. "Development of Multitargeting CRISPR-Cas Antimicrobials for Effective Elimination of *Staphylococcus aureus*." 2025 AIChE Annual Meeting. Boston, MA. 11/3/25. (Poster)
- **Dooley, D.**, Trinh, C. T. "Casper: An Integrated Software Platform for CRISPR Gene Editing of Non-Model Organisms and Microbiomes." 2023 AIChE Annual Meeting. Orlando, FL. 11/8/23.
- **Dooley, D.**, Ryu, S., Edwards, J., Ha, K., Giannone, R., Dien, B., Trinh, C. T. "Harnessing Robustness of Thermophilic *Bacillus Coagulans* for Conversion of Switchgrass Hydrolysates to Designer Bioesters at Elevated Temperatures." 2023 AIChE Annual Meeting. Orlando, FL. 11/6/23. (Poster)

AWARDS AND HONORS

- Volunteer of Distinction May 2026
- Yates Dissertation Fellow Aug. 2025–May 2026
- Tickle College of Engineering Fellowship Oct. 2024
- Outstanding Achievement CBE Graduate Student Award Apr. 2024
- Tickle College of Engineering Fellowship Jul. 2023
- Outstanding Promise CBE Graduate Student Award Apr. 2023
- SARIF Summer Research Graduate Assistantship June 2021
- Tennessee Tech Eminence Award for Outstanding Innovation May 2019
- Dr. John C. McGee and Family Scholarship May 2016–May 2019
- University Academic Scholarship from Tennessee Tech University Aug. 2015–May 2019
- Dean's List at Tennessee Tech University May 2016–May 2019

PROFESSIONAL DEVELOPMENT

- Trinh Lab Mentor 2019–Present
 - Coordinate research efforts of junior researchers in the Trinh Lab.
 - Train students in critical thinking and formulation of creative research projects.
 - Provide technical advice and troubleshooting for researchers' projects.
 - Mentees:
 - ⇒ Hannah Boyd (undergraduate researcher) 2023–2026
 - ⇒ Duc Nguyen (graduate researcher) 2024
 - ⇒ Khanh Ha (undergraduate researcher) 2020–2024
 - ⇒ Nathan Gladden (rotating graduate student) 2023
 - ⇒ Samuel Cohen (undergraduate researcher) 2021–2023
 - ⇒ Stephen Kelley (undergraduate researcher) 2021–2022
 - ⇒ Tulsi Tailor (undergraduate researcher) 2021
 - ⇒ Kishan Tailor (undergraduate researcher) 2020–2021
 - ⇒ Tanner Fry (undergraduate researcher) 2019–2021
 - ⇒ Josh Herman (undergraduate researcher) 2019
- Trinh Lab Purchasing Officer 2020–Present
 - Serve as point of contact for all incoming and outgoing goods in the Trinh Lab.
 - Monitor and organize Trinh Lab inventory, purchasing replacements when necessary.
- Trinh Lab Instrument Manager 2019–Present
 - Manage Opentrons Flex and OT-2 robots, Guava flow cytometer, MinION sequencer, Nanodrop, and gel imager.
 - Perform routine maintenance, calibration, and repair for assigned instruments.
 - Develop standard operating procedures and protocols for assigned instruments.

OUTREACH ACTIVITIES

- Mentor for Student Mentoring and Research Training program (SMaRT) 2022
 - Provided mentorship and guidance to undergraduate SMaRT program participants on graduate-level studies.
 - Instructed participants on ethical research practices involving collection, storage, and communication of scientific data.
 - Taught participants molecular cloning techniques and assisted them in assembling a fluorescence-producing plasmid.
- Mentor for High School Introduction to Engineering Systems for Twelfth Graders (HITES¹²) 2020–2021
 - Gave oral presentations to high school students on Trinh Lab research.
 - Conceived and led educational workshops to introduce participants to Trinh Lab research concepts and methodologies.